AMS 572 Project Report (Group 6)

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Contents

1 Introduction

Nearly two decades since its introduction in 2009, Bitcoin has gained mainstream traction as an investment asset. This digital currency is held by major hedge funds [1], Fortune 500 companies [2], and national banks [3]. Additionally, Bitcoin has gained acceptance as an everyday currency with many retailers now accepting it as payment. Most incredibly, Bitcoin was even declared a legal-tender currency of El Salvador in September 2021 [4]. As Bitcoin has become more accepted, retail investors calmored to buy and trade this new asset. Today, you can hold Bitcoin in digital wallets, investment and retirement accounts [5], as well as purchasing shares of Bitcoin-focused exchange-traded funds (ETFs) [6].

Moreover, the meteoric rise in price per bitcoin has spawned a new class of digital assets called cryptocurrencies. Aside from Bitcoin, these cryptocurrencies follow one of two paradigms. The first paradigm describes coins that focus on improving some aspects of the technology such as energy efficiency, transaction speed, or added utility. These coins utilize their own blockchain networking structure leading to their name as "nativecoins". The second type of coin attempts to capitalize on the speculation and excitement around Bitcoin's price by creating eye-catching imitations. These coins use existing blockchain networks and are often themed around a specific internet trend or personality to garner attention. In online circles, these types of coins have earned the title of "memecoin" as they make no novel improvements to the underlying structure.

1.1 Questions of Interest

Due to its mainstream adoption in traditional financial markets by both institutional and retail investors, does Bitcoin's price follow stock market movements? Is Bitcoin truly a new investment asset, or is its price so deeply entangled with traditional investments that it has lost its novelty? The answer to this question has serious ramifications for portfolio allocations. If Bitcoin's movements are unrelated to the stock market, it can be a valuable tool to diversify portfolios and reduce risk. However, if Bitcoin's price follows the movements of the stock market, its utility as an investment asset mirrors similar stocks in the digital sphere.

How do the price movements of Bitcoin proportionally affect native or memecoins? Are investors more excited about the underlying technology powering digital assets or are they seeking an avenue to wildly speculate? The answer to this question will help unravel the motivation fueling the cryptocurrency boom. This is because of the dual purpose of currency, being a store-of-value and a mode of facilitating transactions. Both of these purposes benefit from the price being predictable and stable. If the memecoins are more sensitive to changes in Bitcoin, it highlights their use as a speculative asset. Likewise, this result highlights the utility of nativecoins and investors bet on the underlying technological improvements.

1.2 Description of the Data

The data for this analysis was imported from Yahoo Finance using the `quantmod` package in `R`. This data contains the date, daily opening price, daily high price, daily closing price, and daily volume for each asset specified. The range of dates collected for this analysis ranges from November 1, 2023 to November 18, 2025. This data was then transformed to isolate the log-daily returns over this time period. Additionally, a dummy variable was added to categorize between the nativecoins and the memecoins. When combining the SPY and BTC data, non-trading days such as holidays and weekends were removed.

2 Background

In this section, we review some basic mathematical machinery with regard to financial time series, and discuss the tools we will be using to perform statistical inference.

2.1 Introduction to Financial Time Series

By a time series we mean a sequence of observations taken at regular time intervals. More formally, a **time series** is simply a sequence $\{x_t\}_{t\in T}$ of real numbers x_t, for $T = \{t_0, t_1, t_2, t_3, \ldots, t_n\}$ some finite indexing set of time increments over a certain period[1]. In the case of finance, our sequence $\{x_t\}_{t \in T}$ (which we may write equivalently as x_t for brevity, where the time index T is implicitly understood) represents the evolution of the price of a particular asset over time. To our notation consistent, we denote the price of a particular asset at the end of a trading day t by P_t .

When we study the evolution of an asset price, it is often useful to measure changes in price by computing the daily **rate of return** $\frac{P_t}{P_{t-1}}$ or, more preferably, the **daily log return** time series r_t of P_t, which is given by the formula

$$r_t = \log\left(\frac{P_t}{P_{t-1}}\right). \tag{1}$$

An example of a financial time series (i.e., the daily return of the S&P 500 stock market index), its daily return & daily log return time series is shown in Figure 1 below.

Figure 1: S&P 500 Daily Returns, Daily Log Returns, and Closing Prices between Nov 1, 2023 and Nov 18, 2025.

[1]In the case of real-world data, this indexing set is finite. However, generally we can make T to be any finite, countably infinite, or even uncountably infinite subset of $\mathbb{R}$ (the real numbers).

In what follows, we will use analyze the price of different cryptocurrencies and financial assets using log returns as in Equation 1.

2.2 Inference for the Correlation between Time Series

Recall that, if X and Y are continuous random variables with $E[X] = \mu_X$ and $E[Y] = \mu_Y$, variances $\text{Var}(X) = \sigma_X^2$ and $\text{Var}(Y) = \sigma_Y^2$, then their **covariance** is given by the formula

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]$$

and subsequently their **correlation** is given by

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

How can we define the same notion of correlation for time series to asses a relationship between them? It turns out, if we are given two time series $\{x_t\}_{t=1}^N$ and $\{y_t\}_{t=1}^N$, we can use the **Pearson correlation coefficient**, defined in Equation 2 below as

$$\hat{\rho} = \frac{\sum_{t=1}^n (x_t - \overline{x})(y_t - \overline{y})}{\sqrt{\sum_{n=1}^n (x_t - \overline{x})^2}\sqrt{\sum_{t=1}^n (y_t - \overline{y})^2}}, \tag{2}$$

where $\overline{x}$ and $\overline{y}$ denote the sample means of $\{x_t\}_{t=1}^n$, $\{y_t\}_{t=1}^n$, respectively. I.e.,

$$\overline{x} = \frac{1}{n}\sum_{t=1}^n x_t, \qquad \overline{y} = \frac{1}{n}\sum_{t=1}^n y_t.$$

When studying the relationship between two time series x_t and y_t, we can use the Pearson correlation coefficient in Equation 2 to construct a hypothesis test of $H_0 : \hat{\rho} = 0$ vs. $H_1 : \hat{\rho} \neq 0$ under the following assumptions:

1. The pairs (x_t, y_t) are independent across time: for $t_1 \in T$ and $t_2 \in T$, (x_{t_1}, y_{t_1}) and (x_{t_2}, y_{t_2}) are independent, and

2. (x_t, y_t) are jointly normally distributed.

Under these assumptions, we use the test statistic

$$T = \hat{\rho}\sqrt{\frac{n-2}{1-\hat{\rho}^2}}, \tag{3}$$

where T follows a t-distribution with $df = n - 2$ degrees of freedom under H_0.

In practice, particularly when dealing with financial data, the assumptions 1 and 2 are often not well satisfied. For starters, the log return of any particular stock often *does not* follow a normal distribution, which violates assumption 2. Furthermore, the development of the price of a stock may not be generally independent across time. However, for purposes of simplification, it is often assumed to be the case in the theory financial modeling that stock prices are independently and identically normally distributed.

2.3 Linear Regression on Financial Time Series

For **Question 2**, we are interested in whether or not the price of memecoins track closer to the price of Bitcoin versus other, more serious nativecoins. We will study this question by performing a linear regression analysis. Let

$$\text{Memecoin}_i = \begin{cases} 1 & \text{if coin } i \text{ is a memecoin,} \\ 0 & \text{if coin } i \text{ is a nativecoin.} \end{cases}$$

We then fit the model

$$\text{Return}_{i,t} = \beta_0 + \beta_1 \text{BTC}_t + \beta_2 \text{Memecoin}_{i,t} + \beta_3 (\text{BTC}_t \times \text{Memecoin}_{i,t}) + \varepsilon_{i,t}. \tag{4}$$

We can interpret the coefficients in Equation 4 in the following way: β_1 can be viewed as the sensitivity of nativecoins to changes in the price of Bitcoin (notice that when $\text{Memecoin}_i = 0$ for a nativecoin i, the slope of Equation 4 is simply β_1). Moreover, β_3 can be viewed as the *difference* in sensitivity for memecoins versus nativecoins, since when $\text{Memecoin}_i = 1$ for a memecoin, the slope is $\beta_1 + \beta_3$. With this in mind, if it turns out that $\beta_3 > 0$, we have that memecoins are more sensitive to the price of Bitcoin than nativecoins are.

In order to apply the linear regression analysis as in Equation 4, which has the general form $Y = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 XZ$, we require the following conditions to be satisfied by X, Z, and the interaction term XZ:

1. The conditional expectation of Y should be linear in the parameters, i.e., $\mathbb{E}[Y \mid X, Z] = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 XZ$,
2. the regressors X, Z, and XZ should be uncorrelated with the error term,
3. no regressor is a linear combination of the others,
4. $\text{Var}(\varepsilon \mid X, Z)$ is constant (this is called **homoskedasticity**),
5. errors are uncorrelated across observations (this means that there is **no autocorrelation**), and
6. ε is normally distributed.

Again, in the case of financial time series, assumptions 4 and 5 are often not satisfied. As a result, if we were to perform a standard linear regression analysis, we would not be able to trust our OLS standard errors and their associated p-values. To remedy this fact, we instead compute a different flavor of standard errors, called **heteroskedasticity and autocorrelation consistent standard errors**, which will provide us with valid inference results even when assumptions 4 and 5 are not satisfied.

To motivate and illustrate this alternative approach, consider the standard linear model

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

for $\mathbf{Y} \in \mathbb{R}^n$ a vector of responses, $\mathbf{X} \in \mathbb{R}^{n \times p}$ a design matrix, where the ith row is denoted X_i,

$\boldsymbol{\beta} \in \mathbb{R}^p$ is the vector of parameters, and $\boldsymbol{\varepsilon} \in \mathbb{R}^n$ is the error vector, where under classical OLS assumptions we assume $\text{Var}(\varepsilon_i) = \sigma^2$, where σ is a **constant**, and $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$. We already know that the standard OLS estimator of $\boldsymbol{\beta}$ is given by

$$\widehat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T \mathbf{Y}$$

which is unbiased provided $[E](\boldsymbol{\varepsilon}) = \mathbf{0}$ and the errors are independent of $\mathbf{X}$. If we want to perform the hypothesis test

$$H_0 : \beta_j = 0,$$

and compute a p-value, we compute the test statistic

$$t_j = \frac{\widehat{\beta}_j}{\text{SE}(\widehat{\beta}_j)} = \frac{\widehat{\beta}_j}{\sqrt{\widehat{\text{Var}}(\widehat{\boldsymbol{\beta}})}} \tag{5}$$

where $\widehat{\text{Var}}(\widehat{\boldsymbol{\beta}}) = \widehat{\sigma}^2 (X^T X)^{-1}$ is the sample variance of the OLS estimator. However, the problem is, this formula for the sample variance is *only* valid when the assumption that the errors are IID holds. In the case where this does not hold (in particular, in the case where the errors are correlated and heteroskedastic), we need to use a different estimator of the covariance matrix of $\widehat{\boldsymbol{\beta}}$, called the ***Newey-West estimator***.

The Newey-West estimator of the covariance matrix of $\widehat{\boldsymbol{\beta}}$ is given by

$$\widehat{\text{Var}}_{\text{NW}}(\widehat{\boldsymbol{\beta}}) = (X^T X)^{-1} \left(X^T \widehat{\Omega}_{\text{NW}} X \right) (X^T X)^{-1}, \tag{6}$$

where $\widehat{\Omega}_{\text{NW}}$ is the complicated expression

$$\widehat{\Omega}_{\text{NW}} = \left(\frac{1}{n} \sum_{t=1}^{n} \widehat{\varepsilon}_t \widehat{\varepsilon}_t \mathbf{X}_t \mathbf{X}_t^T \right) + \sum_{h=1}^{L} \left(1 - \frac{h}{L+1} \right) \left(\frac{1}{n} \sum_{t=h+1}^{n} \widehat{\varepsilon}_t \widehat{\varepsilon}_{t-h} \mathbf{X}_t \mathbf{X}_{t-h}^T + \left(\frac{1}{n} \sum_{t=h+1}^{n} \widehat{\varepsilon}_t \widehat{\varepsilon}_{t-h} \mathbf{X}_t \mathbf{X}_{t-h}^T \right)^T \right).$$

In this case, we replace $\text{SE}(\widehat{\beta}_j)$ in Equation 5 with $\text{SE}_{\text{NW}} = \sqrt{\widehat{\text{Var}}_{\text{NW}}(\widehat{\boldsymbol{\beta}})}$ using Equation 6. The resulting p-values can then be trusted and the multiple linear regression analysis can be performed fully. In `R`, we simply use the `sandwich` and `lmtest` package, fit our linear model `model <- lm(...)` as usual, calculate the Newey-West covariance matrix with `NeweyWest(model)`, and display the regression results using `coeftest(model, NeweyWest(model))`.

2.4 Autocorrelation

In the previous section, it was mentioned that one of the crucial assumptions to apply linear regression analysis is that the errors are uncorrelated across observation, and in the specific case of time series, this is the requirement that there is no **autocorrelation**.

Generally speaking, the autocorrelation of a time series is a measurement of how much a particular time series of interest x_t correlates with a delayed copy of it x_s, where $s \neq t$ and $|t - s|$ is called the ***lag***. We define the ***autocorrelation function*** of a time series x_t as follows:

$$\rho_x(s, t) = \mathbb{E}[x_s \cdot x_t]. \tag{7}$$

Of course, in practice, we have to estimate ρ_x using sample data. In this case, given time series data $\{x_1, x_2, \ldots, x_n\}$ (e.g., n days of closing prices), if we let $\overline{x} = \sum_{t=1}^{n} x_t/n$, and let

$$c_k := \sum_{t=1}^{n-k} \frac{(x_{t+k} - \overline{x})(x_t - \overline{x})}{n},$$

then the ***sample autocorrelation function*** is given by

$$\hat{\rho}_x(k) = \frac{c_k}{c_0}. \tag{8}$$

In this case, the input k is called the ***lag***. We typically abbreviate both the sample autocorrelation function in Equation 8 and the usual autocorrelation function in Equation 7 both as **ACF**.

To assess whether or not a time series has autocorrelation, it is typically useful to plot values of $\hat{\rho}_x(k)$ for $k = 1, \ldots, n$ in what is called a ***correlegram***. A correlegram plots the correlation between the time series and a copy of itself lagged by k in time. When plotting the correlegram of a time series in R, we use the `acf` function (part of the `stats` package). For example, we can plot the ACF of the S&P 500 log returns from Figure 1 in Figure 2 below:

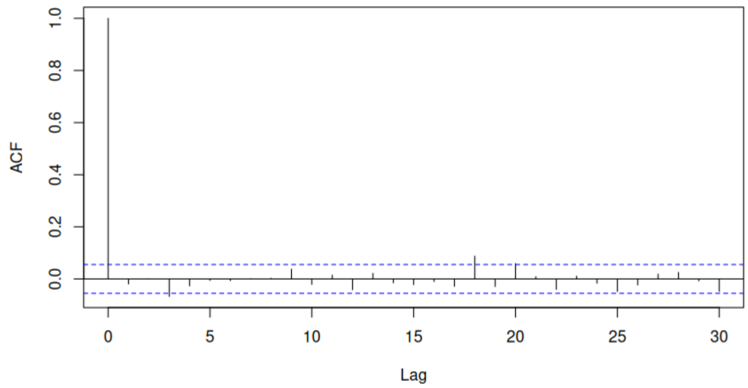

Figure 2: Plot of sample autocorrelation function $\hat{\rho}(k)$ for lag values k.

When plotting the ACF in R, a 95% confidence interval is provided to show whether a correlation between a time series and a copy of itself lagged behind by k is statistically significant. Looking at the ACF plot in Figure 2, we can see a large spike showing $\hat{\rho}(0) = 1.0$, and this is expected. A time series fully correlates with a pure copy of itself with no lag. Looking further out on the plot, we see there is not much self-correlation with the S&P 500 daily log returns and lagged copies of itself, except for small spikes at day 20, 25, and 30 (for instance). Overall, looking at Figure 2, we see that it is reasonable to apply models which require data have no autocorrelation on the S&P 500 log returns.

3 Exploratory Analysis of the Data

To understand Bitcoin's behavior relative to a traditional equity benchmark we compare the growth of a $1 investment in BTC to the S&P 500 ETF (SPY), along with the corresponding daily volatility in Figure 3 below. These comparisons highlight the structural differences between cryptocurrency markets and conventional equity markets.

Figure 3: Growth of $1.00 USD invested in S&P 500 vs. Bitcoin from November 1, 2023 - November 18, 2025

To explore the relationship between daily returns of BTC with nativecoins and memecoins we started with scatterplots of individual coin returns against BTC along with their fitted linear regression lines. For nativecoins, we display this relationship in Figure 4 below.

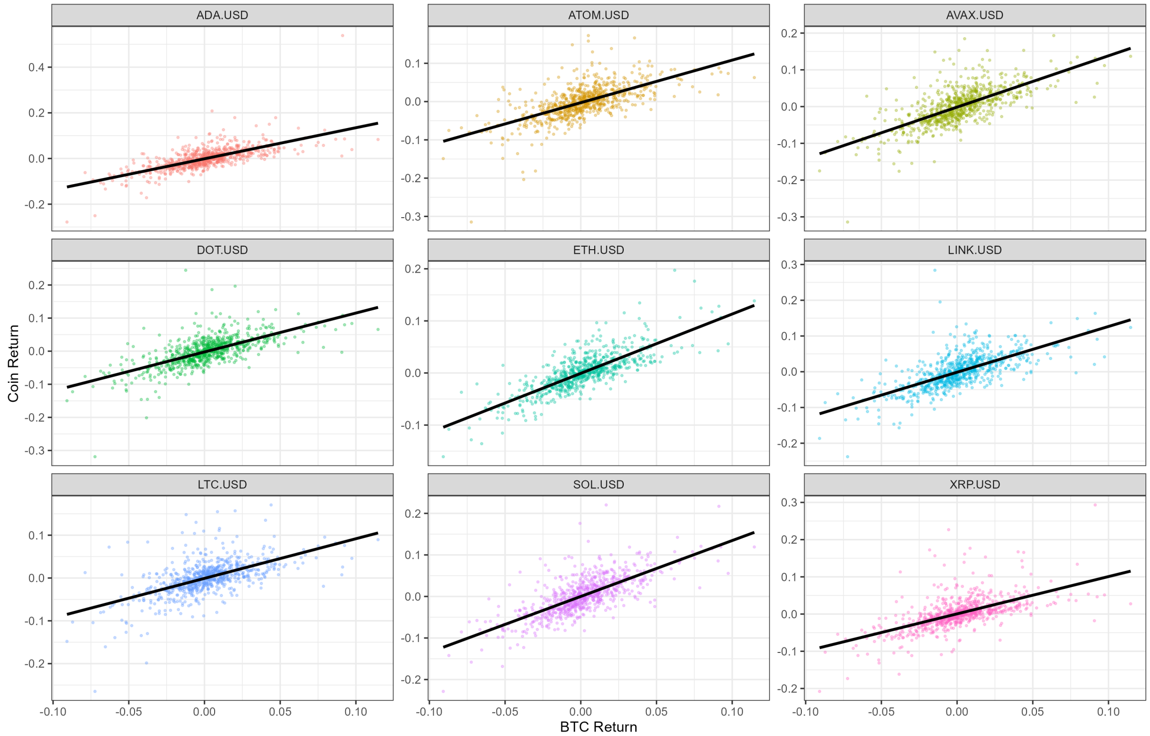

Figure 4: Scatter plots of individual nativecoin returns against BTC along with their fitted linear regression lines.

Across the nativecoins (ADA, ATOM, AVAX, DOT, ETH, LINK, LTC, SOL, XRP), the returns show a strong positive correlation with BTC. The scatterplots show relatively tight clustering around the fitted regression lines, and the slopes are uniformly positive. This suggests that nativecoins respond in a similar way to market movements of BTC. For memecoins, we display their relationship with BTC in Figure 5 below.

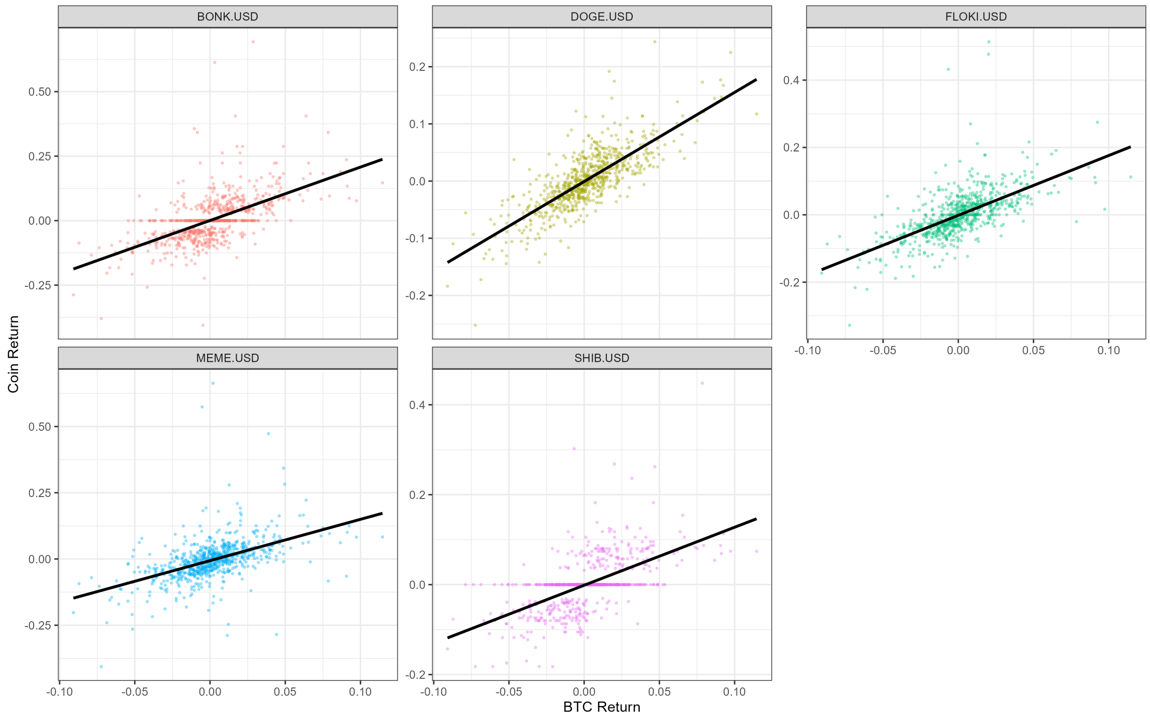

Figure 5: Scatter plots of individual memecoin returns against BTC along with their fitted linear regression lines.

The memecoin group (BONK, DOGE, FLOKI, MEME, SHIB) displays noticeably higher dispersion in returns and weaker alignment with BTC movements. While the fitted slopes remain positive for all tokens, the magnitude of the slopes is generally smaller than that observed for nativecoins, and the returns are more scattered. This suggests that memecoin prices are less attached to the broader crypto-market factor represented by BTC. These tokens appear to exhibit higher volatility. This volatility is often driven by community activity, speculative flows, and social-media-induced demand surges. Even where some memecoins (DOGE) show moderately strong BTC sensitivity, the noise around the fitted line remains substantially larger relative to nativecoins.

Overall this leads us to the idea that nativecoins exhibit stronger, more stable, and more linear sensitivity to BTC, whereas memecoins demonstrate weaker systematic exposure and substantially greater volatility. Nativecoins behave more like assets grounded in platform fundamentals, while memecoins resemble speculative instruments whose return profiles are heavily influenced by other speculative factors rather than market-wide factors. This motivates our deeper dive in question 2.

4 Analysis & Results

4.1 Question 1

4.1.1 Hypothesis Setup and Assumptions

Recall that our first question of study is the following:

Question 1: ***Does Bitcoin's price movements follow those of the S&P 500***?

To formally study this question, we set up the following hypothesis test in Equation 9 below using **Pearson's correlation coefficient** from Equation 2:

$$\boxed{H_0 : \hat{\rho} = 0 \qquad \text{vs.} \qquad H_1 : \hat{\rho} \neq 0} \tag{9}$$

with test statistic from Equation 3, which we recall here for convenience:

$$T = \hat{\rho}\sqrt{\frac{n-2}{1-\hat{\rho}}}$$

We wish to perform this hypothesis test at a **95% confidence level**. The plot in Figure 6 below assesses linearity and is meant to show that model errors have a mean of zero.

In order to perform this hypothesis test, we discuss the fulfillment of key assumptions:

1. **The data is generated from a normal distribution**. This assumption is likely to be violated. This is because financial returns often exhibit fat-tails. To mitigate this behavior of the data, we use the log-returns. This transformation should limit the excess kurtosis. Additionally, since log-returns are additive, the daily log-return is equivalent to the sum of many momentary returns. Due to the efficient market hypothesis[2], these returns result from a random walk. By

[2] The ***efficient market hypothesis*** says, in principle, says that one cannot consistently beat the market. In particular,

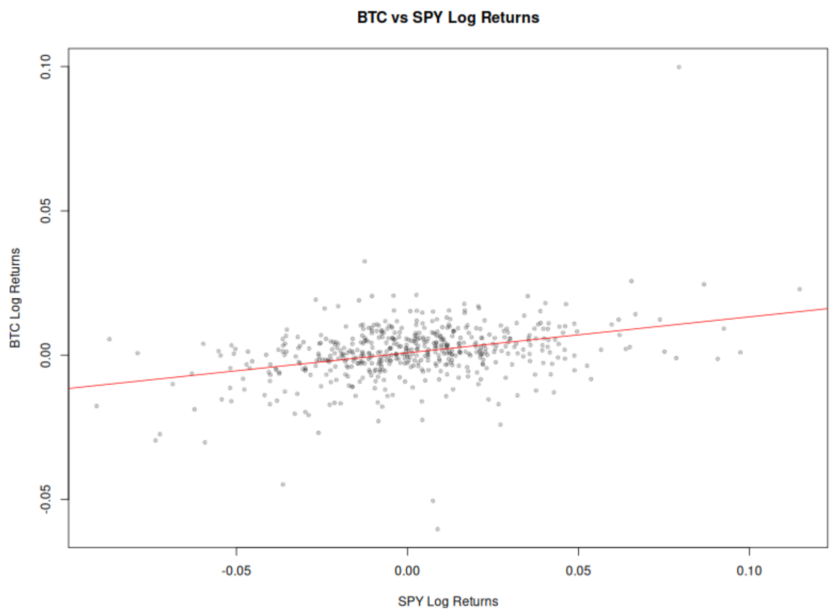

Figure 6: Plot of BTC vs. SPY returns with regression line.

the Central Limit Theorem, this sum should be approximately normally distributed especially when dealing with many data points. This assumption is visually assessed by examining the individual variables' QQ-plots.

2. **There must be a linear relationship between groups.** This again is a reasonable assumption as they are both heavily intertwined with our financial markets. When investors are optimistic about the future, one would expect both the SPY fund and the price of Bitcoin to move. Likewise, when investors are fearful about economic instability, money is generally reallocated away from assets with prospective growth towards assets with current earnings and utility. This assumption can be visually inspected using the scatterplot of the log-returns.

3. **The data is random and from independent samples.** For this time-series data, this means that there is no autocorrelation. This is a reasonable assumption as, due to the efficient market hypothesis, any obvious information from past data, is priced-in to the current valuation. This

the price of a particular asset reflects all available information, so that only new information moves prices. Since new information is unpredictable, it follows that price movements based on that information are unpredictable. Informally speaking, when we sum unpredictable events, we obtain a normally distributed ***random walk***. Therefore, the log return $r_t = \log(P_t/P_{t-1})$, which due to the property of logs can be written as the sum of unpredictable price changes $\log(P_t/P_{t-1}) = \sum_{s=1}^{t} \log(P_s/P_{s-1})$, can be viewed as itself a random walk, which is normally distributed.

assumption is checked using an ACF plot to see if there is significant autocorrelation between the current log-return and lagging log-returns.

The Q–Q plots for SPY and BTC log returns in Figure 7 below reveal important differences and similarities in their distributions relative to a normal distribution. For both assets, the mid-quantiles lie close to the theoretical line. This suggests the bulk of the return distribution is approximately symmetric and only moderately non-normal. However it is apparent that the points deviate from the straight line in both the lower and upper extremes. This indicates that both BTC and SPY returns are more heavy-tailed than a Normal distribution.

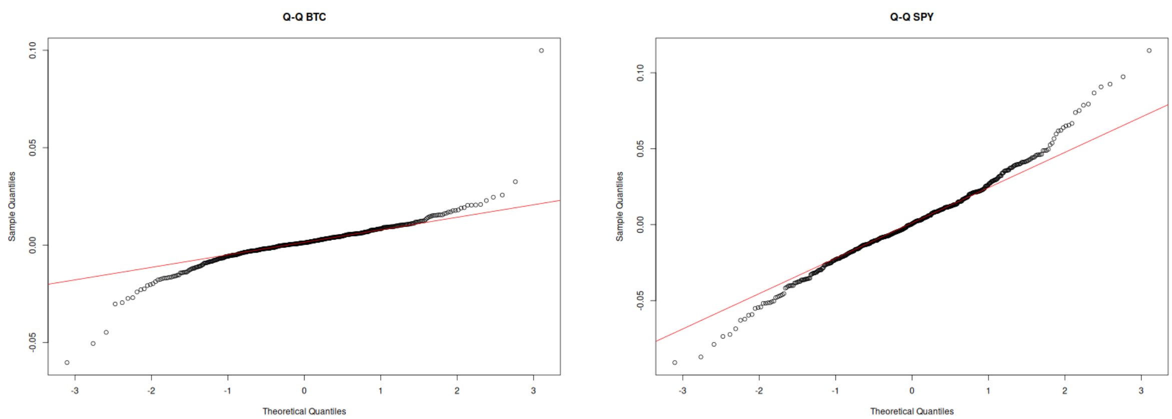

Figure 7: **Left**: Q-Q Plot of BTC returns. **Right**: Q-Q plot of SPY returns.

The BTC Q–Q plot shows a bit more curvature at the extremes, especially on the right tail, where several points rise slightly further from the line. There are one or two returns that stand further above the line than any corresponding point in SPY's Q–Q plot. These are individual observations and do represent a fundamentally different distributional shape

Overall, the Q–Q plots for BTC and SPY look quite similar, showing the typical heavy-tailed structure of financial returns. Both assets depart from normality in broadly the same way. While Bitcoin exhibits slightly stronger tail deviations the difference is modest. This small difference could imply that BTC is slightly more volatile than SPY.

To check assumption 3, we verify that the time series data for BTC and SPY has no autocorrelation. From our discussion back in subsection 2.4, we already illustrated that the S&P 500 log returns do not exhibit autocorrelation (see Equation 8). In the figure below, we verify the same result for Bitcoin:

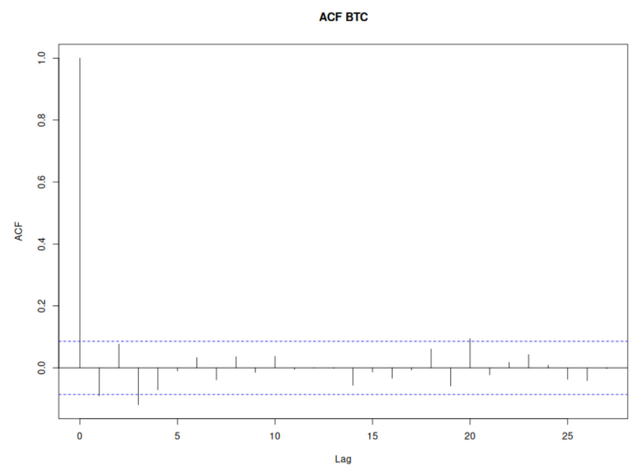

Figure 8: ACF plot for BTC.

As we can see , while there are a couple spikes in correlation at lags of 3 days and 20 days, for the most part, Bitcoin shows no autocorrelation with itself. Therefore, we conclude that our hypothesis test Equation 9 can be reasonably applied to our data.

4.1.2 Question 1 Hypothesis Test Results

Therefore, in R, we coded the hypothesis test in Equation 9 and computed the associated test statistic in Equation 3 and obtained a **Pearson correlation coefficient** of $\hat{\rho} = 0.3499$ and an associated p-value of 2.07×10^{-15}. Since, for our analysis, $\alpha = 0.05 \gg 2.07 \times 10^{-15}$, we **reject the null hypothesis** $H_0 : \hat{\rho} = 0$ and conclude that there is a statistically significant correlation between the daily log returns of Bitcoin and those of the S&P 500.

A Pearson correlation coefficient of $\hat{\rho} = 0.35$ suggests a positive linear relationship, albeit weak, between Bitcoin and the S&P 500.

4.1.3 The Effect of Missing Values for Question 1

In this section, we look at the effect of missing values on the hypothesis test in Equation 9 by deleting S&P 500 return data.

MCAR (Missing Completely at Random). To look at the effect of random missing values in the S&P 500 return data, we deleted 15% of S&P 500 return data in `R` and re-ran the hypothesis test in Equation 9. From this, we obtained (with a fixed randomness seed) a Pearson correlation coefficient of $\hat{\rho}_{\text{MCAR}} = 0.3637$ with a p value of 1.99×10^{-14}, which is still far below our α-threshold of 0.05. Hence, even with 15% of the data missing, we **still reject the null hypothesis**.

The results of our MCAR stress test our reassuring. Given that many firms use the internet to access live market data, packet loss can cause some of that data to never be received. Therefore, the results of our MCAR test show that even with a sizable portion of missing data, the results of our statistical inferences still hold.

MNAR (Missing Not at Random). To look at the effect of non-random missing values in the S&P 500 return data, instead of deleting a random 15% batch of return data, we instead **deleted 10% of the bottom 10% of S&P 500 returns**. One might wonder when this would happen in real life. In the case of a market crash, many investors will be sending in sell orders to liquidate their stock and cut their losses. Such flooding of orders over the internet can potentially cause a server overload and subsequent crash to online stock trading platforms so that traders do receive little, if any, data about stock prices during a crash.

So, in the case of this MNAR analysis, our re-estimation of the Pearson correlation coefficient $\hat{\rho}_{\text{MNAR}}$ when 80% of the bottom 10% of S&P 500 returns are deleted was computed to be $\hat{\rho}_{\text{MNAR}} = 0.3037$ with a p-value of 5.43×10^{-11}. While this is still far smaller than our α-level of 0.05, it is **4 orders of magnitude** greater than our base p-value for the hypothesis test on the full data. So, while we still reject H_0 strongly at the 95% confidence level, the higher p-value still matches with our understanding of how BTC tracks with the S&P. Sharp rises or falls for a given asset happening in sync with another asset is, on its own, strong evidence of correlation between those two assets. Therefore, if one asset experiences a sharp drop, while there is no data (being missing or otherwise) showing a sharp drop during that same instance in time, there will be that much less evidence of correlation between the two.

4.2 Question 2

Recall that our first question of study is the following:

Question 2: ***Do memecoins follow the price changes of Bitcoin more than nativecoins?***

To answer this question, we use the multiple linear regression model with interaction that was first shown in Equation 4:

$$\text{Return}_{i,t} = \beta_0 + \beta_1 \text{BTC}_t + \beta_2 \text{Memecoin}_{i,t} + \beta_3 (\text{BTC}_t \times \text{Memecoin}_{i,t}) + \varepsilon_{i,t}. \tag{10}$$

where we recall that

$$\text{Memecoin}_i = \begin{cases} 1 & \text{if coin } i \text{ is a memecoin,} \\ 0 & \text{if coin } i \text{ is a nativecoin.} \end{cases}$$

Then, for each coin (indexed by $j = 1, \ldots, 14$, 9 nativecoins, 5 memecoins), we compute the test statistic from Equation 5:

$$t_j = \frac{\hat{\beta}_j}{\sqrt{\widehat{\text{Var}}_{\text{NW}}(\widehat{\boldsymbol{\beta}})}}$$

where we recall that $\widehat{\text{Var}}_{\text{NW}}$ is the **Newey-West estimator** of the covariance matrix of $\widehat{\boldsymbol{\beta}}$ given in Equation 6.

Now, with regard to Equation 10, our hypothesis test is as follows:

$$\boxed{H_0 : \beta_3 = 0 \qquad \text{vs.} \qquad H_1 : b_3 > 0} \tag{11}$$

which we test at the **95% confidence level**.

To run this test, we must make the following assumptions about our data:

1. There must be linearity in the parameters. This is a reasonable assumption as all cryptocurrencies involved share an asset class. This assumption can be visually checked in the Residuals vs. Fitted plot.

2. There must be homoskedasticity in the data. This assumption is also likely to be violated due to how financial data exhibits periods of higher and lower volatility. Additionally, memecoins often rely on social-media driven marketing to attract new traders. This is likely to cause periods of much higher volatility over the short term. To mitigate this violation, we will use heteroskedasticity-robust standard errors as discussed. This assumption is visualized in the Scale-Location plot.

3. There must be independence of the errors. This means that there cannot be any autocorrelation. As stated before, this is reasonable to assume due to the efficient market hypothesis.

4. The errors must be normally distributed. The distribution of the residuals are likely to showcase fat-tail behavior similar to the returns. This assumption is visually checked in the QQ-Residuals plot.

Let's start with checking the first hypothesis, being that there is no linearity in the parameters. For this, we have the Residuals vs. Fitted values plot in Figure 9 below:

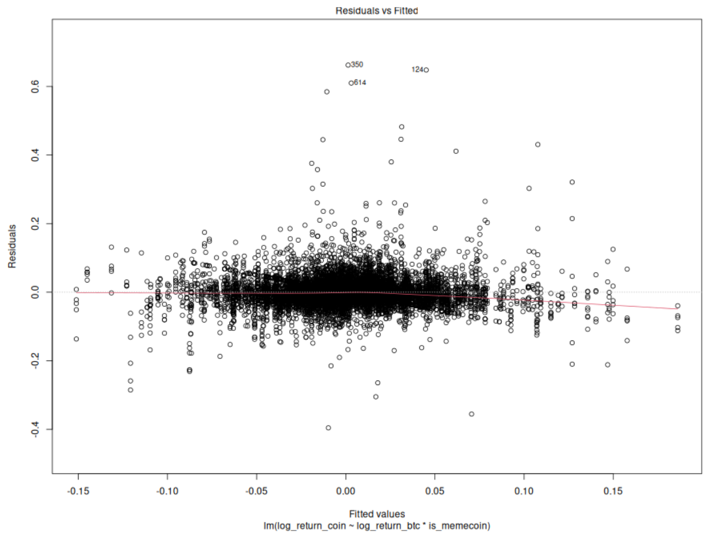

Figure 9: Residuals vs. Fitted Plot.

From the above plot, we see that there is no major systematic curvature in the data (see the red line). Therefore, we can safely say assumption 1 is satisfied.

We already expect from the outset that the homoskedasticity assumption that is required for legacy multiple linear regression will not hold for our financial data, therefore requiring us to use the Newey-West heteroskedasticity-robust standard errors as discussed above. We visualize the lack of homoskedasticity in the *scale-location* plot below, which illustrates the relationship between the fitted values and the variability of the residuals:

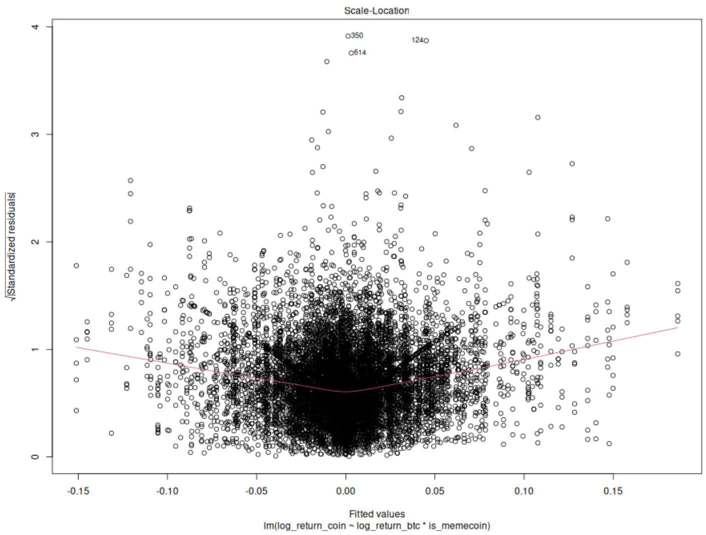

Figure 10: Scale-Location plot of fitted values vs. residuals.

As we can see, there is a strong V-shape in the red line in Figure 10 above, indicating our data is heteroskedastic, which reinforces our motivation to use the Newey-West approach to multiple linear regression.

Next, we check to see if the errors exhibit autocorrelation, which we expect. While this expectation is a violation of one of the crucial assumptions, the Newey-West approach is robust to autocorrelation, and hence the ACF plot of the errors showing strong autocorrelation in Figure 11 below further justifies our use of the Newey-West standard error formulation.

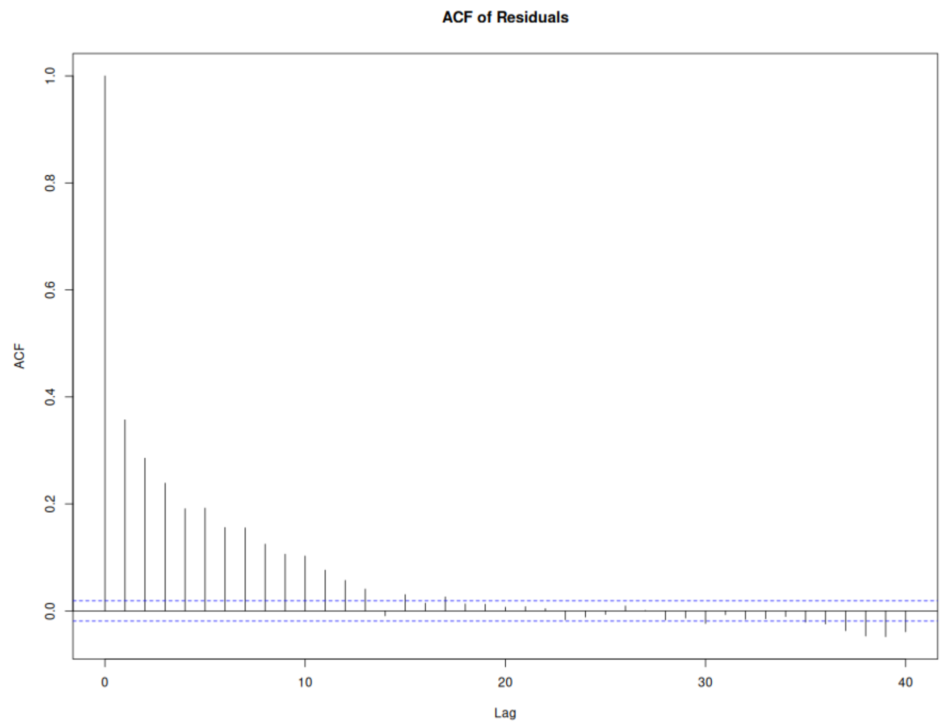

Figure 11: ACF plot of residuals.

Finally, we check for normality of the distribution of errors. While we expect them to show fat tails, as these are financial returns, our MLR is still valid as our Newey-West approach does not require errors to be normally distributed. Again, we are justifying our approach using Newey-West standard errors.

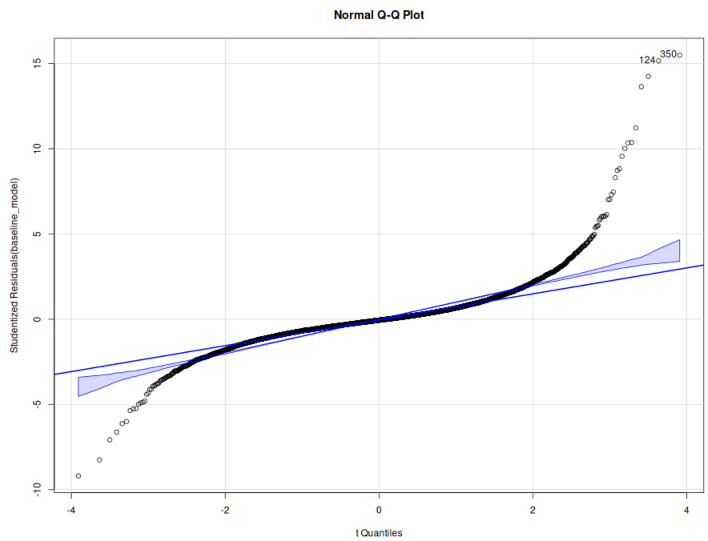

Figure 12: QQ plot of errors.

As seen in Figure 12 above, the QQ-plot shows that the errors exhibit fat tails, but otherwise follow a linear alignment towards the center.

4.2.1 Question 2 Hypothesis Test Results

Therefore, after coding the hypothesis test in Equation 11, we obtained a value for $\beta_3 = 0.4627$ at a p value of 4.96×10^{-10}. Therefore, we **strongly reject the null hypothesis** $H_0 : \beta_3 = 0$ and conclude that there is very strong statistical evidence suggesting that memecoins are more heavily levered on Bitcoin than nativecoins are.

4.2.2 The Effect of Missing Values for Question 1

MCAR (Missing Completely At Random). To look at the effect of random missing values on our hypothesis test in Equation 11, we wanted to see what would happen to our hypothesis test results of we randomly deleted 20% of memecoin returns. Memecoins, as their name suggests, are just patently not serious coins, and can have weak infrastructure making data transmission unreliable. Deleting 20% of memecoin returns can reflect a simple packet loss in data transmission over the internet, say, between a user and the crypto trading platform.

So, in the case of this MCAR test, we obtained a value of $\beta_{3,\text{MCAR}} = 0.4515$ at a p-value still less than 2.77×10^{-9}. So, even in the case of deleting 20% of memecoin return data, we still strongly reject the null hypothesis. This makes sense when looking at our untampered hypothesis test. The correlation between memecoins and Bitcoin is so strong, that we can conclude that memecoins are, compared to nativecoins, proportionally more sensitive to price changes in bitcoin. So, even if we deleted a continuous window of data, we would still see this effect.

MNAR (Missing Not At Random). In terms of the effect of non-random missing values in the crypto data, we decided to take a similar approach to our MNAR analysis of first question. Here, we first wrote `R` code to identify the lowest 5% of memecoin returns, and delete 90% of those observations. Again, this is reflective of a real-world event where holders of memecoins ”panic sell”, perhaps on news that the owner of that particular memecoin is about to ”pull the rug”[3]. In this case, the results of our hypothesis test change somewhat significantly, and we end up with a coefficient of $\beta_{3,\text{MNAR}} = 0.2319$ at a p-value of 0.00213. So, in this case, while we still are *just* able to reject H_0, our new coefficient value says that memecoin is only ***half*** as levered as in the original hypothesis test (which gave us $\beta_3 = 0.4627$). However, our p value is much larger. If we were to require 99.9% confidence, this hypothesis test would fail. Conceptually, this makes sense, for similar reasons as discussed in the MNAR analysis for question 1: the leverage of one asset on another is strongly revealed when one asset crashes and another follows suit simultaneously; if that data is removed, the effect will no longer be observed. In this MNAR analysis, we are targeting an event (a crash in the memecoin price) that is most likely to reveal heavy leverage on BTC or not. Now, despite a lower β_3 value, we still see that outside of crashes memecoins are strongly levered on BTC, more than nativecoins.

[3]”Pulling the rug” in crypto refers to when the developers of a particular coin suddenly liquidate all of the coin’s value. This crashes the price to zero, and all other investors lose all of their money

5 Conclusion

In this report, we examined two questions dealing with cryptocurrencies. The first question was whether or not Bitcoin's daily log returns had similar price movements as those of the S&P 500. What we found was a statistically significant correlation, albeit somewhat weak, between BTC and the S&P 500. This suggests that Bitcoin is not wholly independent of traditional markets. Our MCAR and MNAR tests that, even with the loss of data, this statistically significant correlation remains.

The second question was a study into whether ”meme” coins respond more strongly to Bitcoin's price movements as compared to traditional, well-known ”native” coins. We found the answer to this question to be in the affirmative after running a multiple linear regression with interaction. To our surprise, not only was the result statistically significant, but it was *strongly* so, as even with MCAR and MNAR tests we obtained extremely small p-values for the estimates of the regression coefficients.

Our approach is not without its limitations. As noted above, the time-series nature of this financial data requires us to relax assumptions of the models employed. Despite our best efforts in mitigating these violating characteristics, there will always be some level of autocorrelation and non-stationarity in this data. On this note, the data range of two years represents a small sliver of time relative to the long history of financial markets, which has undergone many fundamental shifts. Markets are ever-changing, and the results of this analysis may not have external validity when generalizing to future behaviors under different market conditions. A significant limitation of this study is survivorship bias associated with the selection of assets. New cryptocurrencies are created constantly, many of which have a short lifespan. These coins, often dubbed “sh-t coins,” never garnered enough prominence to reach our study. In a similar vein, there have been nativecoins that have lost all of their value. One prominent example is the FTX token of Sam Bankman-Fried's FTX cryptocurrency exchange. This token has lost 99% of its value from its peak valuation and is at risk of being completely liquidated to pay creditors during the ongoing bankruptcy proceedings.

This selection bias highlights the deviations in risk between coins that is unrelated to the movements of other coins. Additionally, it stresses the need for more study in determining what other factors (market capitalization, founder's social media presence, etc.) influence the price movements of cryptocurrencies as well as more empirical and academic ways to classify subgroups of this asset class.

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